

Vermillion: Oil Painting Simulation In Virtual Reality

A new tool for digital artists offering the analog control of traditional painting with the benefits of a virtual environment.

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ABSTRACT

Digital art has come a long way since the first release of MS Paint. And while artists can already work wonders with the tools at their disposal, there's always been a very distinct barrier between traditional mediums and the digital ones. Instead of brushes on a canvas, a stylus on a screen is used. Instead of mixing paints of different colors and thickness, a color picker and transparency slider are presented. The instinctiveness and versatility of bristles on canvas have been replaced with a selection of discrete tools - draw, blur, smudge,... It's also not possible to make strokes with large gestures, to work on your piece from arm's length, like the old masters did. The typical texture that physical paint has is also lost in drawing programs. *Vermillion* offers a new tool for digital artists, be they novices who always wanted to try their hand at following along with Bob Ross, or veteran Photoshop users who long for more analog control. It simulates the full oil painting experience. A selection of specialist brushes is offered, each with unique characteristics and their distinct brush pattern. The bristles bend as they are pushed against a surface, soaking up the paint that's on the palette. The paints mix as they would in the real world, not as RGB colors do on a monitor. Paint thinner can be used to control the thickness and flow of the paint. The paint can be pushed, pulled, blurred and blended on the canvas, and shows the relief and pattern of your strokes. The resulting piece can be exported to be presented as a new digital work of art.

CCS CONCEPTS

• **Applied computing** → Arts and humanities; Fine arts; Computers in other domains; Personal computers and PC applications; Computer games.

KEYWORDS

Virtual Reality, Simulation, Oil Painting

ACM Reference Format:

Thomas van den Berge . 2021. Vermillion: Oil Painting Simulation In Virtual Reality: A new tool for digital artists offering the analog control of traditional painting with the benefits of a virtual environment.. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Immersive Pavilion (SIGGRAPH '21 Immersive Pavilion)*, August 09–13, 2021, Virtual

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SIGGRAPH '21 Immersive Pavilion, August 09–13, 2021, Virtual Event, USA

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ACM ISBN 978-1-4503-8368-4/21/08.

<https://doi.org/10.1145/3450615.3464542>

Event, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3450615.3464542>

1 INTRODUCTION

As you start the application and strap on your headset, you enter your virtual studio. Directly in front of you stands your easel. In one hand, you're holding a large palette, in the other a brush. Next to the easel is a brush rack, presenting you with all the tools you have at your disposal for creating your first piece. Switching brushes is as straight forward as grabbing one off the rack. On the other side of the easel is a web browser, booting up YouTube. It offers you the sheer endless range of painting tutorials on the internet, or the ability to look for reference. Starting your work of art is as easy as dipping your brush into some paint and stroking it across the canvas.

2 DESIGN AND IMPLEMENTATION

2.1 User experience

The design goal was to create a true pick-up-and-play experience. Modern painting programs can overwhelm a first time user with a daunting amount of toolbars and panels; *Vermillion* invites the user to just use their hands instead. When holding a brush and a palette, you will instinctively try to get paint on the brush by moving one onto the other. A small haptic feedback buzz is triggered when the brush touches a surface, letting the user know that contact has been made. Making a mark or stroke of a certain size does not require changing brush size settings, but simply holding the brush at a different angle.

2.2 Paint loading and mixing

You load the bristles with color by smooshing them into the paint on the palette. If you touch the paint lightly, only a little bit of paint is visibly loaded on the brush, and vice versa. You mix colors by simply brushing through piles of paint of different color. Color mixes subtractively in *Vermillion*, for instance allowing you to mix green from yellow and blue. In addition to paint, the brush can also be doused with paint thinner, making the brush appear wet. An important aspect of oil painting is the concept of "thick over thin", i.e. you start with very thin paint which can easily be painted over with thicker paint. Running paint of equal or lower thickness through each other makes the colors blend - a useful tool for creating smooth transitions in between two values. The simulated paint thickness in *Vermillion* also allows for impasto effects, i.e. the use of thicker paint on the canvas to create the desired visual, next to the colors themselves.

3 COMPARISON TO EXISTING APPLICATIONS

There are several other two dimensional VR painting applications already available. Infectious Ape's *Kingspray* has been available since 2016, but rather than painting it simulates graffiti art. Oisoi's *Painting VR* has become available in early access in 2021, and mainly focuses on painting on large canvases. While it offers some of the same functionality as *Vermillion* such as simulated bristles and the ability to blur paint on the canvas, it does not offer a realistic paint mixing experience, taking a more digital approach instead. *Brushwork VR* and *VARTISTE* are a web based VR painting tools, and do not offer realistic paint mixing, simulated bristles, impasto effects, or pulling and blurring paint on the canvas.

4 DISCUSSION

As it stands, *Vermillion* already offers a unique experience in Virtual Reality, giving artists a new way of expressing themselves using the tools that created the world's masterpieces brought into the digital realm. There are still plenty of opportunities for the application to grow, however. It is currently set inside a virtual studio, giving you a browser as a window to the world. But real value to the painting process and user experience could be added by instead immersing the painter in a complete virtual landscape, allowing her or him to choose a location and time of day to plant their easel and paint what they see around them. Or for portraiture, by adding photorealistic 3D models of humans which can be costumed, posed and lit as desired. Next to being a tool for novice and professional artwork creation by individuals, the application could also be used to offer remote painting lessons by bringing together students and teachers in the same virtual environment.